

Magnetic Fields — Mark schemes

PAPERS A TO D • 30 MARKS EACH • TUTOR COPY

Marks in square brackets show how the total is split. This topic is marked on wording, not on arithmetic, so award marks for the specific statement rather than for the general idea. Almost every mark lost goes one of four ways: assuming every metal is magnetic; field lines drawn without arrowheads or with the arrows reversed; treating attraction as proof that something is a magnet; and confusing the geographic North Pole with a magnetic north pole. The notes under each answer flag the specific mistake that question was built to catch.

Paper A — Medium

30 MARKS

- 1 Two from: iron, nickel, cobalt, steel [1] [2]
Two from: copper, aluminium, brass, gold, silver, zinc, lead, tin [1]
The second half must be metals. Answering plastic and wood misses the entire point of the question and scores zero for that mark.
- 2 iron yes • copper no • nickel yes • aluminium no • cobalt yes • brass no [3]
All six correct [3]; four or five correct [2]; two or three correct [1]
Brass is the one that separates them. It is copper and zinc, so it is not magnetic — students who guess from the word "metal" get it wrong, and students who know the three magnetic elements get it right.
- 3 (a) They repel (push apart) [1] [3]
(b) They repel [1]
(c) They attract [1]
(b) catches anyone who has memorised "north repels north" as a fact about north rather than about like poles.
- 4 A magnetic material is one that a magnet attracts, such as iron, nickel, cobalt or steel [1] [2]
A magnet is a magnetic material that has been magnetised, so it has its own poles and produces its own field [1]
Every magnet is made of a magnetic material; almost no magnetic material is a magnet. Accept any wording that keeps those two apart.
- 5 (a) At least four smooth lines running from one pole round to the other [1] [4]
Arrowheads on every line, pointing north to south outside the magnet [1]
Lines closer together at the poles than at the sides, and never crossing [1]
(b) Away from the north pole [1]
Award the shape mark generously and the arrow mark strictly. A beautiful pattern with no arrows scores 1 out of 3 here, which is exactly the lesson.
- 6 Arrows on every line, running from north to south outside the magnet [1] [3]
Lines never cross [1]
Lines are closer together where the field is stronger (at the poles) [1]
Also accept: lines form complete loops / lines leave and enter the pole faces at right angles (true at the ends of the bar, not along its sides)
"Draw them neatly" and "use a pencil" are not rules and score nothing. Three physics statements are wanted.
- 7 Closer lines mean a stronger field, so the field is strongest at the poles [1] [2]
The spacing is the only part of the diagram that carries the strength information, so even spacing states something untrue [1]

- 8** Place the compass near one pole and mark a dot at each end of the needle [1] [3]
 Move the compass so its tail sits on the previous head dot, mark the new head, and repeat to the other pole [1]
 Join the dots into a smooth curve and add an arrow in the direction the compass north end pointed [1]
The arrow is part of the method. A described procedure that ends with a bare curve has recorded the shape and thrown away the direction — cap at 2.
- 9** Magnetism produced in a magnetic material by placing it in a magnetic field [1] [2]
 It is temporary: the near end becomes the opposite pole so the object is attracted, and the magnetism is lost when the field is taken away [1]
- 10** Soft iron magnetises easily but loses it at once; steel is harder to magnetise but keeps it [1] [2]
 Soft iron: electromagnet cores (temporary magnets). Steel: bar magnets and compass needles (permanent magnets) [1]
The pairing gets reversed under pressure. Soft iron with temporary, steel with permanent — reversing it loses both marks, not one.
- 11** The Earth has a magnetic field, and the needle is a small magnet free to turn [1] [2]
 The needle's north-seeking pole is attracted towards the magnetic pole in the far north (which is a magnetic south pole), so it lines up along the field [1]
- 12** The test is repulsion: only a magnet can be repelled by a magnet [1] [2]
 Attraction is not enough because an unmagnetised magnetic material is attracted to either pole by induced magnetism [1]
This appears in some form on all four papers because it is the single most reliable source of dropped marks in the topic.

Paper B — Medium

30 MARKS

- 1** (a) Iron nail, steel spoon and nickel disc [1] and no others [1] [4]
 (b) Copper wire and the aluminium can are both metals and were not attracted [1]
 so being a metal does not make something magnetic; only iron, nickel, cobalt and their alloys are [1]
The nickel disc is the one that gets left off (a). Pure nickel is magnetic — it is the third element, and it is forgotten more often than cobalt.
- 2** Steel is mostly iron, and iron is magnetic [1] [3]
 Brass is an alloy of copper and zinc [1]
 Neither copper nor zinc is magnetic, so the brass key is not attracted [1]
Award the third mark only if the answer names why brass fails, rather than simply asserting that it does.
- 3** (a) Four or more smooth lines from pole to pole [1]; arrowheads on every line running north to south [1]; closer spacing at the poles, no crossings [1] [5]
 (b) Two from: arrowheads present and pointing the right way; lines not crossing; spacing tighter at the poles; lines starting and ending on the magnet [1] [1]
- 4** Compass placed near a pole, dots marked at each end of the needle [1] [3]
 Compass moved so the tail sits on the last head dot, repeated across to the other pole [1]
 Dots joined into a smooth curve with an arrow added in the direction the compass north end pointed [1]

- 5 A crossing point would mean the field there pointed in two directions at once [1] [3]
 which is impossible, since a compass placed there settles in one direction only [1]
 The two lines should run side by side without meeting [1]
Answers that only say "field lines never cross" have restated the rule, not explained it. That is worth one.
- 6 (a) The magnet's field induces magnetism in the steel clip [1]; the near end becomes a south pole, and unlike poles attract [1] [4]
 (b) The first clip is now itself a (temporary) magnet [1], so it induces magnetism in the second clip in the same way [1]
Watch for "the magnetism travels down the chain". Nothing travels — each clip is magnetised in turn by the field of the one above it.
- 7 (a) A magnetic south pole [1], because the needle's north-seeking pole is attracted to it and unlike poles attract [1] [4]
 (b) The Earth's field exerts forces on both poles of the magnet, turning it [1] until its north-seeking pole lines up along the field, pointing north; there is then no turning effect left [1]
"Because the Earth is a magnet" alone scores one in (a). The justification mark needs the words unlike poles attract.
- 8 (a) An unmagnetised magnetic material becomes a temporary magnet by induction [1], and is attracted to either pole, so attraction only proves it is magnetic material [1] [4]
 (b) Bring the object near one pole of a known magnet, then the other [1]; if it is repelled by either, it is a magnet [1]

Paper C — Hard

30 MARKS

- 1 (a) The bar is not a magnet — it is unmagnetised magnetic material [1] [5]
 (b) The magnet's field induces magnetism in the steel [1]; whichever pole is offered, the near end of the bar becomes the opposite pole, so it is attracted either way [1]
 (c) If the bar were a magnet it would have been attracted by one pole and repelled by the other [1]; the absence of any repulsion is what rules that out [1]
Two attractions look like two pieces of evidence and are worth one between them. The candidate has to notice that the informative observation — a repulsion — never happened.
- 2 Try all three pairs of ends and look for attraction [1] [4]
 The bar attracted by nothing at all is the brass [1]
 Of the remaining two, touch the end of one to the middle of the other [1]
 Strong attraction there means the bar being held is the magnet, because the middle of a magnet has almost no field of its own [1]
Repulsion is unavailable here — only one bar is magnetised, so nothing ever repels. Candidates who insist on the repulsion test and stop have applied a correct rule to the wrong situation; award the first two marks.
- 3 (a) They push apart [1]; two north poles are like poles and like poles repel [1] [5]
 (b) At the neutral point, midway between them, 3 cm from each [1]; the two fields there are equal in size and opposite in direction, so they cancel and the resultant field is zero [1]
 (c) $6 \div 2 = 3$, and $3^2 = 9$ times stronger [1]
In (c) the common wrong answer is 3, from forgetting to square. The midpoint answer in (b) is only correct because the magnets are identical — say so if the candidate does.

- 4 (a) Away from the magnet (to the left) [1] [4]
 (b) Towards the magnet (also to the left) [1]
 (c) Horizontally, to the right — from the north pole towards the south pole [1]
 (d) Above the magnet the field lines run across from the north pole to the south pole, and the compass north end lies along the field [1]
(a) and (b) both point left, which is why they are set together. Beyond the north pole the field leaves the magnet; beyond the south pole it returns to it. Candidates who answer "left" and "right" have reasoned from the picture rather than the field.
- 5 (a) Lifts the steel cans and iron girders; leaves the aluminium cans and copper wire [1]; [4]
 because only iron and its alloys are magnetic [1]
 (b) Copper and aluminium are both non-magnetic [1], so the magnet ignores both equally and cannot tell them apart [1]
This is the "all metals are magnetic" trap dressed as an application question. Answers that lift the aluminium cans lose both marks in (a).
- 6 (a) The field inducing each clip is weaker the further it is from the magnet [1], so the [4]
 magnetism induced in each clip is weaker than in the one above it [1]
 (b) The chain stays up [1]; copper is not magnetic and does not block a magnetic field, which passes straight through non-magnetic materials [1]
(b) is the discriminating question on this paper. Most candidates predict the chain falls, reasoning that the copper is in the way. Award the first mark only for a correct prediction with no explanation.
- 7 (a) The buried magnet's south end is at the geographic North Pole [1]; a compass [4]
 needle's north-seeking pole is attracted to it and unlike poles attract [1]
 (b) Useful: it predicts the shape of the field and which way a compass points anywhere on the surface [1]
 Not literally true: the core is far too hot to stay permanently magnetised — the field is produced by moving molten iron, and it drifts over time [1]
Accept any sensible limitation in (b), including that the magnetic poles move and do not sit exactly at the geographic poles.

Paper D — Hard

30 MARKS

- 1 (a) In a line, with like poles facing each other across the gap — north facing north, or [5]
 south facing south [1] [1]
 (b) A point where the fields of the two magnets are equal and opposite, so they cancel and the resultant field is zero [1]
 (c) It moves towards the weaker magnet [1]; you have to get closer to the weaker one before its field has grown enough to match the stronger one's [1]
(c) separates the candidates who have memorised "halfway" from the ones who understand what is cancelling. Halfway is only right for identical magnets.
- 2 (a) Touch the end of one bar to the middle of the other [1] [5]
 Strong attraction: the bar being held is the magnet [1]
 Little or no attraction: the other bar is the magnet, and repeating the test the other way round confirms it [1]
 (b) The middle of a magnet has almost no field, because the poles are at the ends [1]; an unmagnetised bar has no poles anywhere, so its middle behaves the same as its ends [1]
With only two bars and one magnet, nothing ever repels — the repulsion test is unavailable and candidates must find another route. Reward any test that uses the weak middle of a magnet.

- 3** Lines stopping in mid-air: every line must run from the magnet's surface back to it, forming a closed loop [1] [4]
 Even spacing: spacing shows strength, so the lines must be closer together at the poles [1]
 Arrows into the north pole: outside the magnet the arrows run north to south, so they should point out of the north pole [1]
 Lines touching: field lines never cross or touch, since the field has one direction at each point [1]
- 4** (a) Steel keeps its magnetism, so a compass needle stays magnetised permanently [1]; soft iron magnetises and demagnetises easily, so an electromagnet can be switched off [1] [4]
 (b) Store them in pairs with unlike poles together, and soft iron keepers across the ends [1]; this gives the field a closed path through magnetic material, so the magnet does not gradually demagnetise itself [1]
Storage is often not taught and often examined. Accept "keep them away from heat, hammering and dropping" for one mark in (b) with a correct reason.
- 5** (a) The compass points to magnetic north, not geographic north [1]; the magnetic pole is not in the same place as the geographic pole, so the two directions differ by an angle [1] [4]
 (b) The steel gate is magnetised by induction in the Earth's field [1]; the Earth's field is very weak, so close to the gate the gate's own field is stronger and the needle lines up with that instead [1]
(b) rewards anyone who has grasped how weak the Earth's field actually is. It is also why a plotting-compass practical says to clear iron off the bench.
- 6** (a) The filings collect at the two ends — the poles [1]; that is where the field is strongest [1] [4]
 (b) Two shorter magnets, each with its own north and south pole [1]; a single isolated pole cannot be produced, however many times you break it [1]
"One half is north and the other half is south" is the wrong answer here and is chosen often. It scores nothing.
- 7** (i) The mistake: attraction only shows the clip is magnetic material, not a magnet [1] [4]
 Correct version: the steel clip is temporarily magnetised by induction; only repulsion would prove it was a magnet [1]
 (ii) The mistake: outside the magnet the lines run north to south, not south to north [1]
 Correct version: reverse every arrow, and draw the lines closer together at the poles rather than evenly spaced [1]
Both statements are the two most common errors in the topic, written out as another student's work. Diagnosing them is the same cognitive job as avoiding them, and much easier to face.